

POSITIVE GENERALIZED CORNER-VECTOR DESIGNS ON $\mathbb{S}^3$ HAVE DEGREE AT MOST 13

ABSTRACT. Tanino, Tamaru, Hirao, and Sawa have proved that positive weighted spherical designs obtained from generalized corner-vector orbits in dimension four have degree at most 15, and asked whether a weighted 15-design on $\mathbb{S}^3$ with more than five proper orbits exists. We give a B_4 -invariant polynomial of degree 14 that is positive on every generalized corner-vector orbit but has negative spherical mean. Consequently, every positive design of this type on $\mathbb{S}^3$ has degree at most 13. This gives a negative answer, for positive weights, to their Problem 6.1, independently of the number of proper orbits.

1. THE OBSTRUCTION

Let B_4 act on $\mathbb{R}^4$ by signed coordinate permutations. For $a > 0$ and $s \in \{0, 1, 2, 3\}$, set

$$v_{a,s} = \frac{1}{\sqrt{a^2 + s}}(a, \underbrace{1, \dots, 1}_s, 0, \dots, 0) \in \mathbb{S}^3,$$

and denote its B_4 -orbit by $v_{a,s}^{B_4}$. These are the generalized corner-vector orbits of Tanino–Tamaru–Hirao–Sawa [1, Definition 2.19]; the orbit is called proper when $a \neq 1$. They proved that a positive weighted design assembled from such orbits has degree at most 15 in dimension four [1, Theorem 3.1], and they asked [1, Problem 6.1]:

Does there exist a weighted 15-design on $\mathbb{S}^3$ of type (1) with more than 5 proper orbits?

Their “type (1)” is the display numbered (1) on p. 1388 of [1]: a cubature formula for normalized surface measure whose node set is a union of generalized corner-vector orbits, carrying one weight per orbit. That is the class of formula (1) below, and it is also the class occurring in the hypothesis of [1, Theorem 3.1], whose formula (11) has the same shape. Let σ denote normalized surface measure on $\mathbb{S}^3$.

Theorem 1. *Let $m \geq 1$, $a_i > 0$, $s_i \in \{0, 1, 2, 3\}$, and $W_i > 0$. The formula*

$$(1) \quad \int_{\mathbb{S}^3} f d\sigma = \sum_{i=1}^m W_i \sum_{x \in v_{a_i, s_i}^{B_4}} f(x)$$

cannot be exact for every polynomial f of degree at most 14.

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Corollary 2. *Every positive weighted spherical design of generalized corner-vector type on $\mathbb{S}^3$ has degree at most 13. In particular, Problem 6.1 of [1] has a negative answer for positive weights, as required of a weighted design by [1, Definition 2.2], regardless of the number of proper orbits.*

Put $y_i = x_i^2$, and let $e_j = e_j(y_1, y_2, y_3, y_4)$ be the j th elementary symmetric polynomial. Define

$$(2) \quad p = 2 + 240e_2 + 1345e_3 + 227924e_4 - 3518e_2^2 - 448e_2e_3 + 283604e_3^2 \\ - 1529173e_2e_4 + 12886e_2^3 + 4855151e_3e_4 - 132972e_2^2e_3.$$

Then p is B_4 -invariant and has degree 14 in $x_1, \dots, x_4$. For a polynomial f on $\mathbb{S}^3$, write

$$\langle f \rangle = \int_{\mathbb{S}^3} f d\sigma.$$

Lemma 3. *The spherical mean of p is negative:*

$$\langle p \rangle = -\frac{60763}{143360} < 0.$$

Proof. For a uniformly distributed point of $\mathbb{S}^3$, the vector (y_1, y_2, y_3, y_4) has the Dirichlet distribution with parameters $(1/2, 1/2, 1/2, 1/2)$. Hence, for $\alpha_i \in \mathbb{Z}_{\geq 0}$,

$$\langle y_1^{\alpha_1} \dots y_4^{\alpha_4} \rangle = \frac{\prod_{i=1}^4 (1/2)^{\alpha_i}}{(2)_{\alpha_1 + \dots + \alpha_4}}, \quad (z)_0 = 1.$$

Expanding (2) and applying this identity gives the stated value. $\square$

Lemma 4. *For every $a > 0$ and $s \in \{0, 1, 2, 3\}$,*

$$p(v_{a,s}) > 0.$$

Proof. For $s = 0$, one has $v_{a,0} = (1, 0, 0, 0)$ and $p(v_{a,0}) = 2$. Let $s \in \{1, 2, 3\}$ and set

$$u = \frac{a^2}{a^2 + s} \in (0, 1), \quad c = \frac{1 - u}{s}.$$

Since p is a symmetric function of the y_i , we may take the squared coordinates of $v_{a,s}$ to be $(u, c, \dots, c, 0, \dots, 0)$, with c repeated s times. Direct substitution into (2) gives

$$p(v_{a,1}) = H_1(u), \quad 32p(v_{a,2}) = H_2(u), \quad 729p(v_{a,3}) = H_3(u),$$

where

$$H_1(u) = -12886u^6 + 38658u^5 - 42176u^4 + 19922u^3 - 3758u^2 + 240u + 2, \\ H_2(u) = -598374u^7 + 2387827u^6 - 3979288u^5 + 3540713u^4 \\ - 1750294u^3 + 440357u^2 - 42268u + 1391, \\ H_3(u) = -51606520u^7 + 213893327u^6 - 355558770u^5 + 299081978u^4 \\ - 130946506u^3 + 26917938u^2 - 1819702u + 39713.$$

We certify positivity by exact Bernstein coefficients. For an interval $I = [r, q] \subseteq [0, 1]$ and a polynomial H of degree d , write

$$H(r + (q - r)z) = \sum_{k=0}^d \beta_{I,k}(H) \binom{d}{k} z^k (1 - z)^{d-k}.$$

If $H(r + (q - r)z) = \sum_{j=0}^d c_j z^j$, then

$$(3) \quad \beta_{I,k}(H) = \sum_{j=0}^k c_j \frac{\binom{k}{j}}{\binom{d}{j}}.$$

For each s , the following table gives a subdivision $0 = r_0 < r_1 < r_2 < r_3 = 1$ and the minimum Bernstein coefficient on each successive interval, where $d = \deg H_s$, so that $d = 6$ for $s = 1$ and $d = 7$ for $s = 2, 3$. The coefficients (3) depend on d ; in particular H_1 is not to be padded to degree 7.

s	(r_0, r_1, r_2, r_3)	$(\min_k \beta_{[r_{j-1}, r_j], k}(H_s))_{j=1}^3$
1	$(0, \frac{1}{6}, \frac{5}{6}, 1)$	$(\frac{14287}{7776}, \frac{46531}{23328}, \frac{14287}{7776})$
2	$(0, \frac{1}{12}, \frac{3}{4}, 1)$	$(\frac{137627845}{2985984}, \frac{414113333}{5971968}, \frac{219941}{5376})$
3	$(0, \frac{1}{20}, \frac{3}{5}, 1)$	$(\frac{542097724127}{448000000}, \frac{17317137}{15625}, \frac{4479262}{3125})$

Every displayed number is positive. Since the Bernstein basis functions are nonnegative on $[0, 1]$ and sum to 1, it follows that $H_s(u) > 0$ on every listed interval, hence on $[0, 1]$. Therefore $p(v_{a,s}) > 0$ for $s = 1, 2, 3$ as well. $\square$

Proof of Theorem 1. Assume that (1) is exact through degree 14. Applying it to the polynomial p and using its B_4 -invariance gives

$$-\frac{60763}{143360} = \langle p \rangle = \sum_{i=1}^m W_i |v_{a_i, s_i}^{B_4}| p(v_{a_i, s_i}).$$

The right-hand side is strictly positive by $W_i > 0$ and Lemma 4, contradicting Lemma 3. $\square$

Remark 5. Since $-I \in B_4$, every formula of the form (1) is antipodal. Exactness through degree 14 is therefore equivalent to exactness through degree 15.

Positivity is essential, and the positive-weight reading is the one intended in [1]: a weighted design there carries by definition a weight function with values in $\mathbb{R}_{>0}$ [1, Definition 2.2], so the hypothesis $W_i > 0$ of Theorem 1 is theirs and not a restriction added here. The displays [1, (1) and (11)] do not repeat the sign condition; that the positive reading is nonetheless the intended one is confirmed by [1, §6], where it is recorded, immediately before Problem 6.1 is posed, that 15-designs on $\mathbb{S}^3$ with negative weights do exist. Theorem 1 accordingly says nothing about signed cubature formulas,

nor about node families outside the generalized corner-vector construction. Degree 13 is attained within this class: [1, §6] reports two weighted 13-designs on $\mathbb{S}^3$ of this type, each with five orbits and positive weights, with nodes and weights given there numerically to six significant figures. On the evidence of those examples the bound of Corollary 2 is sharp; verifying them exactly is not attempted here.

Exact verification. Everything above is finite and exact, and is recomputable from the data printed here; the only mathematical input is the eleven coefficients of (2). From them one expands p into its 130 monomials in the y_i and confirms its B_4 -invariance and its degree 14 in $x_1, \dots, x_4$; evaluates $\langle p \rangle = -60763/143360$ by two routes, direct expansion against the Dirichlet moment functional of Lemma 3 and a harmonic route through the Laplacian, the moment functional itself being cross-validated against the 24-cell through degree 5 and against an iterated-Laplacian formula on all 330 even monomials of degree at most 14; substitutes the profiles of Lemma 4 to recover H_1, H_2, H_3 with their integrality scalings 1, 32, 729; verifies the identity (3) itself, rather than assuming it, and with it all nine minimum Bernstein coefficients of the table; and re-certifies positivity of H_1, H_2, H_3 on $[0, 1]$ a second way, by exact Sturm root counting. No floating-point arithmetic enters at any stage. This computation was re-run independently in exact rational arithmetic, and all 48 of its checks agree with the numbers printed here. The program and its output log are retained in an auxiliary archive, available on request; nothing above depends on them, since the Bernstein table is itself the certificate.

Scope. Two limits belong with that statement. First, the claim quantifies over the continuum $a > 0$, so no census is possible and none was attempted: the universal statement is discharged exactly by the Bernstein certificate over the tabulated partition, re-certified by Sturm, with exact evaluation of p at 609 genuine orbit points as a further check. Second, agreement is what is claimed for the independent re-run, not a second proof: a few of its checks are self-tests of its own construction, which no object could fail, and the agreement of the Pochhammer and double-factorial forms of the Dirichlet moment is an algebraic rewriting rather than an independent derivation.

REFERENCES

- [1] K. Tanino, T. Tamaru, M. Hirao, and M. Sawa, More on the corner-vector construction for spherical designs, *Algebraic Combinatorics* **8** (2025), no. 5, 1387–1414, doi:10.5802/alco.441.